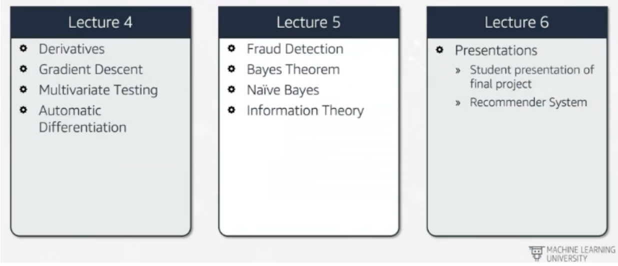
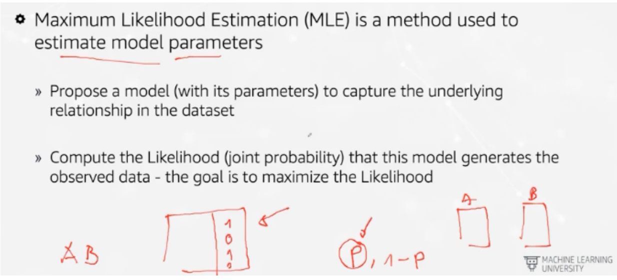
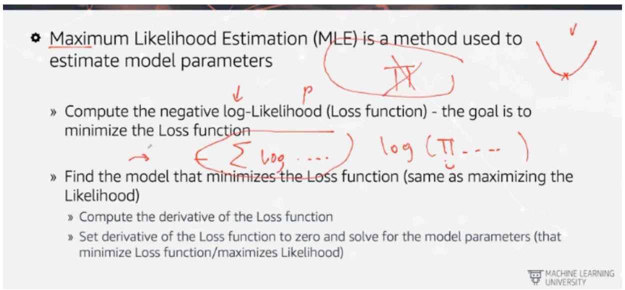
Bayes and Information Theory

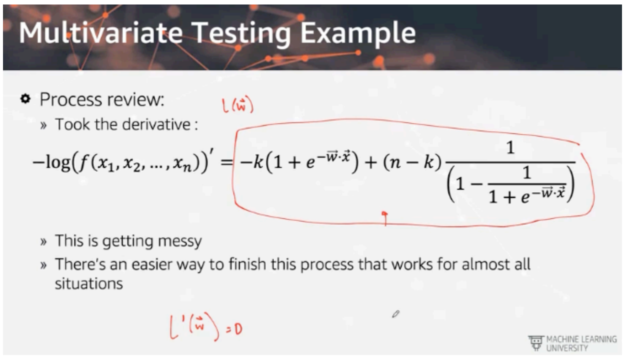
Sunday, 18 February 2024 08:56

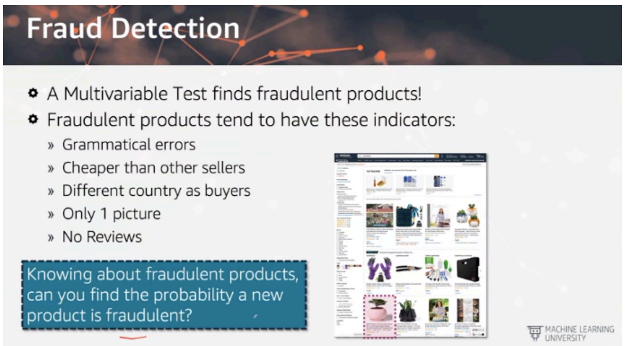


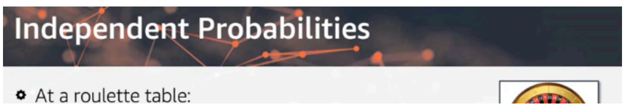
De lo que se trata es de encontrar la función que Maximiza la probabilidad de volver a encontrar un dataset como los que estamos comparando, y urilizar esa función para estimar los nuevos valores que se nos planteen.

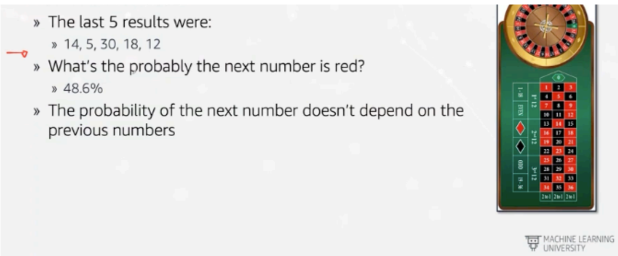


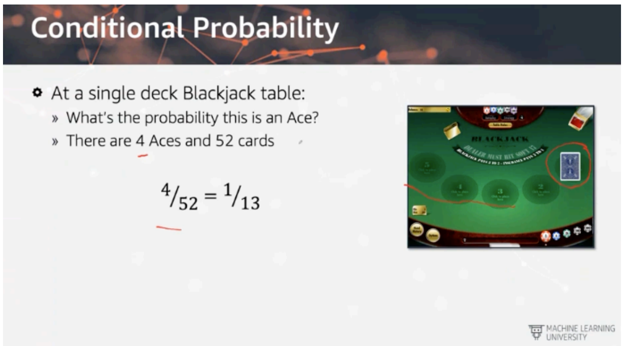


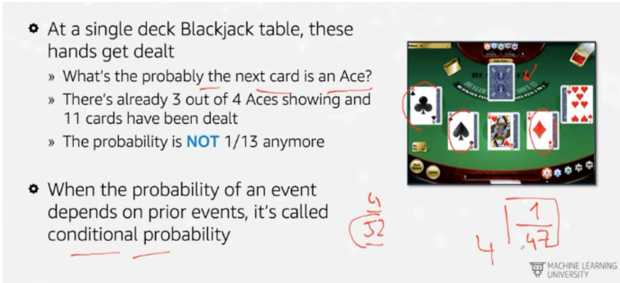


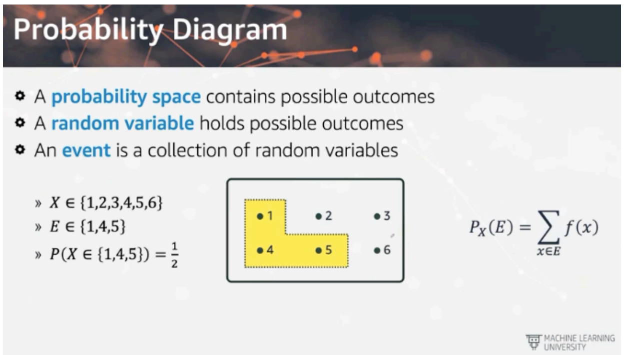




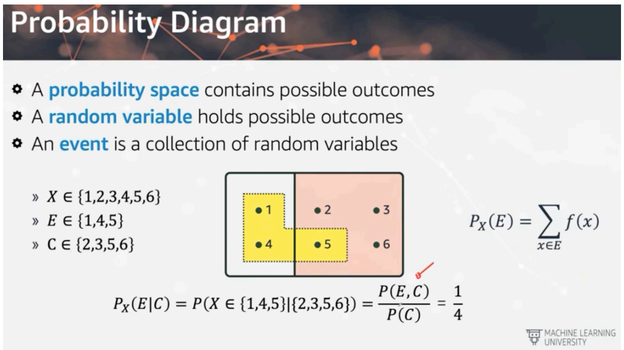


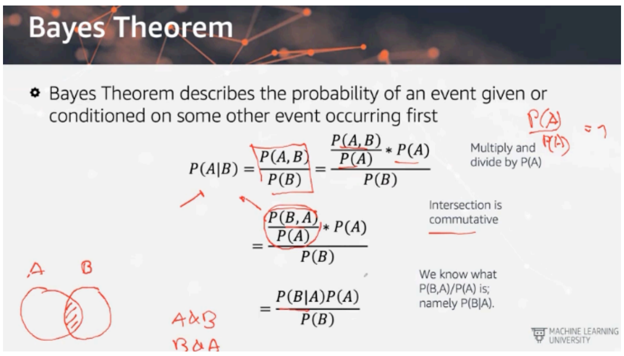


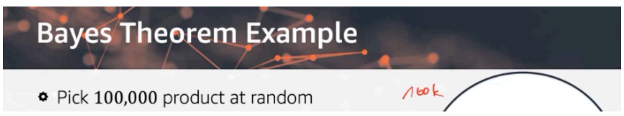




La manera de introducir probabilidad condicional es restringiendo el espacio de probabilidad con el que se puede contar, por ejemplo aquí abajo cual es la probabilidad sabiendo que sólo están los números 2,3,5 y 6 (solo hay 1 de los 4 que podría ser), estoy condicionando mi evento.



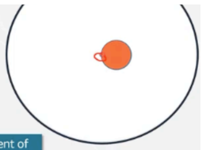




» $^{1}/_{1000}$ will have fraud

» $^{9}/_{10}$ of those with fraud will be flagged

» $^{1}/_{100}$ will be flagged for fraud when they are not fraudulent



What percentage of flagged products are fraudulent?

Starting number of products	Number of fraudulent products	Number of fraudulent products that are flagged	Number of products that are not fraudulent	Number of products that are not fraudulent but are flagged	Percent of flagged products that are actually fraudulent
100,000	100	90	99,900	999	90/(90+999)

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Naïve Bayes

- Can Bayes Theorem to find fraud?
- Looking at data shows fraudulent products tend to have these indicators:
  - » Grammatical errors
  - » Cheaper than other sellers
  - » Different country from buyers
  - » Only one pictures
  - » No reviews

$$P(Fraud|Grammar, Cheaper, Country, Picture, Reviews)$$

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Naïve Bayes

- Start with a sample of products and aggregate to one variable:

Product ID	Grammar	Cheapest	Country	Picture	Review	Fraudulent
hi3993jsk	1	0	1	0	0	0
wp9913ioe	0	0	0	0	0	0
9f9923ek	0	1	0	0	0	0
jfkeeOwi	0	0	0	0	0	0
89kj21333	1	1	1	0	1	1
p2pio134	0	0	0	0	0	0
oiu123i3i	0	1	0	0	0	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮
nil389sna	0	0	0	0	0	0
nilkd2890	1	1	1	0	0	1
jil892kd	0	0	0	0	0	0
hjile389sj	0	0	1	0	0	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮
8uhb3efv	0	0	0	0	0	0
wd2efv96t	1	1	1	1	1	1
⋮	⋮	⋮	⋮	⋮	⋮	⋮

	Fraud	Not Fraud
Grammar	3	3
Not Grammar	0	25

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Naïve Bayes

- Marginal distribution of counts

	Fraud	Not Fraud	Total
Grammar	3	3	6
Not Grammar	0	25	25
Total	3	28	31

Naïve Bayes

- Joint probability distribution

	Fraud	Not Fraud	Total
Grammar	3/31=9.7%	3/31=9.7%	6
Not Grammar	0/31=0.0%	25/31=80.6%	25
Total	3	28	31

- Marginal probability distribution

	Fraud	Not Fraud	Total
Grammar	3	3	6/31=19.4%
Not Grammar	0	25	25/31=80.6%
Total	3/31=9.7%	28/31=90.3%	31

La probabilidad de que no pase se representa con el apóstrofe (no significa derivada ni nada por el estilo)

Naïve Bayes

- Marginal probability distribution

P(F | . . . .)

	Fraud	Not Fraud	Total
Grammar	3	3	6/31=19.4%
Not Grammar	0	25	25/31=80.6%
Total	3/31=9.7%	28/31=90.3%	31

P(G)

P(C

P(not F)

Naïve Bayes

- Conditional probability distribution

	Fraud	Not Fraud	Total
Grammar	3	3/28=10.7%	6
Not Grammar	0	25/28=89.3%	25
Total	3	28	31

$$P(G|F') = 10.7\%$$
$$P(G'|F') = 89.3\%$$

Naïve Bayes

- Conditional probability distribution

	Fraud	Not Fraud	Total
Grammar	3/6=0.5	3/6=0.5	6
Not Grammar	0/25=0	25/25=1	25
Total	3	28	31

$$P(F|G) = 0.5$$
$$P(F'|G) = 0.5$$
$$P(F|G') = 0$$
$$P(F'|G') = 1$$

Naïve Bayes

- Getting conditional probability distribution from marginal and joint probability

Conditional probability

	Fraud	Not Fraud	Total
Grammar	3/6=50%	3	6
Not Grammar	0	25	25
Total	3	28	31

Joint probability

	Fraud	Not Fraud	Total
Grammar	3/31	3/31	6/31
Not Grammar	0	25	25
Total	3	28	31

Marginal probability

$$P(F|G) = \frac{P(G,F)}{P(G)} = \frac{3/31}{6/31} = \frac{3}{6} = 50.0\%$$

Vamos a hacer el mismo cálculo, pero en lugar de tener un solo B (grammar) vamos a tomar todas las variables

Naïve Bayes

- Can Bayes Theorem to find fraud?

$$P(A|B) = \frac{P(A,B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

$$P(F|\underbrace{G, Ch, Co, P, R}_{B}) =$$

$$= \frac{\boxed{P(G, Ch, Co, P, R|F)}P(F)}{P(G, Ch, Co, P, R)}$$

Product ID	Grammar	Cheapest	Country	Picture	Review	Fraudulent
hi3993jsk	1	0	1	0	0	0
wp9913ioe	0	0	0	0	0	0
9f9923ek	0	1	0	0	0	0
jfkeeOwi	0	0	0	0	0	0
89kj21333	1	1	1	0	1	1
p2pio134	0	0	0	0	0	0
oiu123i3i	0	1	0	0	0	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮
nil389sna	0	0	0	0	0	0
nilkd2890	1	1	1	0	0	1
jil892kd	0	0	0	0	0	0
hjile389sj	0	0	1	0	0	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮
8uhb3efv	0	0	0	0	0	0
wd2efv96t	1	1	1	1	1	1
⋮	⋮	⋮	⋮	⋮	⋮	⋮

- Can Bayes Theorem to find fraud?

$$P(F|G, Ch, Co, P, R) =$$

$$= \frac{\boxed{P(G, Ch, Co, P, R|F)}P(F)}{P(G, Ch, Co, P, R)}$$

  - Do not always have enough data for every possible row permutation
    - Make a "Naïve" assumption
    - Assume columns are independent

Product ID	Grammar	Cheapest	Country	Picture	Review	Fraudulent
hi3993jsk	1	0	1	0	0	0
wp9913ioe	0	0	0	0	0	0
9f9923ek	0	1	0	0	0	0
jfkeeOwi	0	0	0	0	0	0
89kj21333	1	1	1	0	1	1
p2pio134	0	0	0	0	0	0
oiu123i3i	0	1	0	0	0	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮
nil389sna	0	0	0	0	0	0
nilkd2890	1	1	1	0	0	1
jil892kd	0	0	0	0	0	0
hjile389sj	0	0	1	0	0	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮
8uhb3efv	0	0	0	0	0	0
wd2efv96t	1	1	1	1	1	1
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Naïve Bayes

- **Independent** variables do not have an effect on each other's probabilities
  - Ways to mathematically define independence:

$$P(A|B) = P(A) \qquad P(A,B) = P(A|B)\,P(B)$$

$$\frac{P(A|B)}{P(A)} = 1 \qquad \boxed{P(A,B) = P(A)\,P(B)}$$

Naïve Bayes

- Naïve Assumption
  - Assume the variables, once Fraud is held constant, are independent

$$P(G, Ch, Co, P, R|F) = P(G|F)P(Ch|F)P(Co|F)P(P|F)P(R|F)$$

- This assumption doesn't always hold
  - Might be a reason Naïve Bayes underperforms when compared to other methods

Naïve Bayes

- Can Bayes Theorem to find fraud?

$$P(F|G, Ch, Co, P, R) = \frac{P(G, Ch, Co, P, R|F)P(F)}{P(G, Ch, Co, P, R|F)P(F) + P(G, Ch, Co, P, R|F')P(F')}$$

  - The Naïve assumption gives enough to calculate the numerator
  - What about the denominator?

$$P(A) = P(A|B)P(B) + P(A|B')P(B')$$

- Can Bayes Theorem to find fraud?

$$P(F|G, Ch, Co, P, R) = \frac{P(G, Ch, Co, P, R|F)P(F)}{P(G, Ch, Co, P, R|F)P(F) + P(G, Ch, Co, P, R|F')P(F')}$$

  - The Naïve assumption gives enough to calculate the numerator
  - What about the denominator? Use independence assumption too.

$$P(F|G, Ch, Co, P, R) = \frac{P(G|F)P(Ch|F)P(Co|F)P(P|F)P(R|F)P(F)}{P(G|F)P(Ch|F)P(Co|F)P(P|F)P(R|F)P(F) + P(G|F')P(Ch|F')P(Co|F')P(P|F')P(R|F')P(F')}$$

Ahora se transforma en un ejercicio purmente de contar de nuevo para cada variable

- Calculating Bayesian Classifiers

	Fraud	Not Fraud	Totals
Different country	3	7	10
Same country	0	21	21
Totals	3	28	31

$$P(Co|F) = \frac{P(Co, F)}{P(F)} = \frac{3/31}{3/31} = 1 \qquad P(Co|F') = \frac{P(Co, F')}{P(F')} = \frac{7/31}{28/31} = 0.25$$

Naïve Bayes

- What the probability of a new product being fraudulent?

Product Id	Grammar	Cheapest	Country	Picture	Review	Fraudulent
7d944oew	1	1	1	0	0	$P(F\|G = 1, Ch = 1, Co = 1, P = 0, R = 1)$

$$P(F|G = 1, Ch = 1, Co = 1, P = 0, R = 0) =$$

$$\frac{P(G = 1|F)P(Ch = 1|F)P(Co = 1|F)P(P = 0|F)P(R = 0|F)P(F)}{P(G = 1|F)P(Ch = 1|F)P(Co = 1|F)P(P = 0|F)P(R = 0|F)P(F) + P(G = 1|F')P(Ch = 1|F')P(Co = 1|F')P(P = 0|F')P(R = 0|F')P(F')} =$$

$$\frac{(1)(1)(1)(0.66)(0.33)(0.097)}{(1)(1)(1)(0.66)(0.33)(0.097) + (0.11\%)(0.25)(0.25)(0.93)(1)(0.9)} = \mathbf{0.792} = \mathbf{79.2\%}$$

Una vez que detectemos la probabilidad de un nuevo producto, el siguiente paso sería establecer un trashold sobre el cual avisar a un anotador humano que lo revise, por ejemplo, todo lo que sea por encima del 75% que sea revisado.

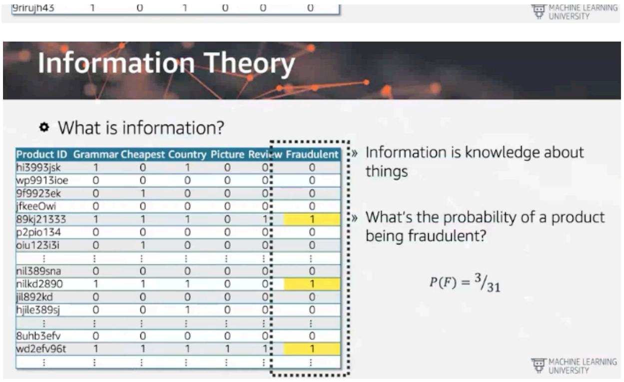
La información se puede definir como la capacidad de reducir la no certeza (incertidumbre)

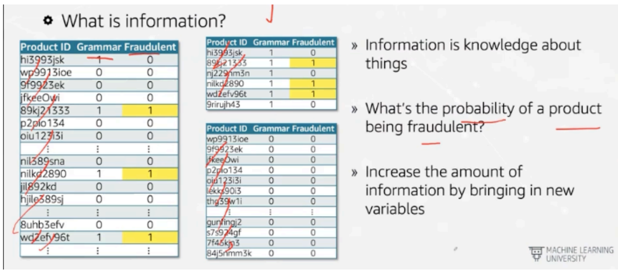
Information Theory

- What is information?

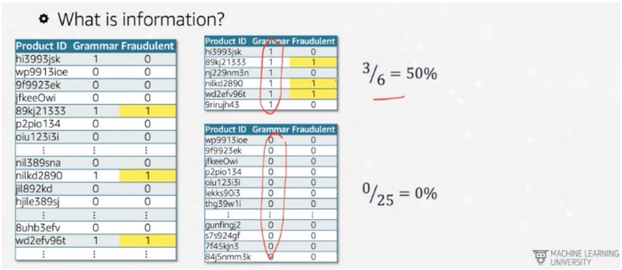
Product ID	Grammar	Cheapest	Country	Picture	Review	Fraudulent
hi3993jsk	1	0	1	0	0	0
wp9913ioe	0	0	0	0	0	0
9f9923ek	0	1	0	0	0	0
jfkeeOwi	0	0	0	0	0	0
89kj21333	1	1	1	0	1	1
p2pio134	0	0	0	0	0	0
oiu123i3i	0	1	0	0	0	0
lekks90i3	0	0	1	0	0	0
thg39w1i	0	1	0	0	0	0
jfik39e3o	0	0	0	0	0	0
nj229nm3n	1	1	0	1	0	0
093jmp22	0	0	0	0	0	0
838wkm32	0	1	1	0	0	0
ijl3ois983	0	0	0	0	0	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

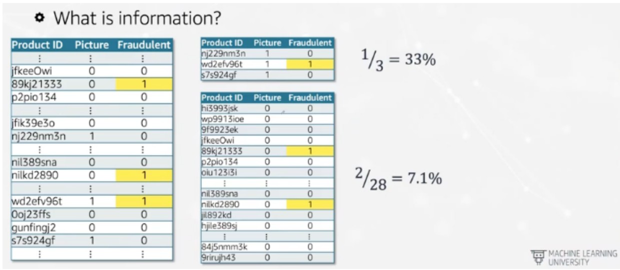
- Information can be thought of as the **resolution of uncertainty**
- Information theory is based on probability theory and statistics
- Information theory often concerns itself with **measures of information of the distributions associated with random variables**

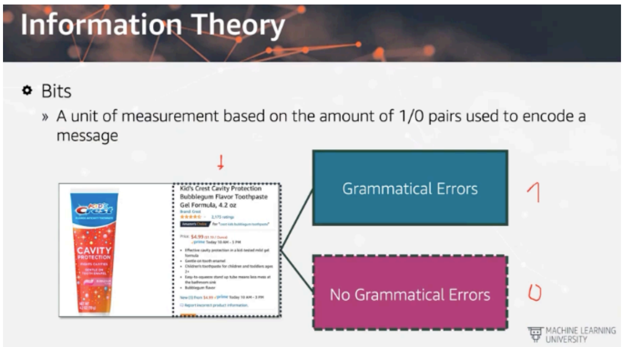


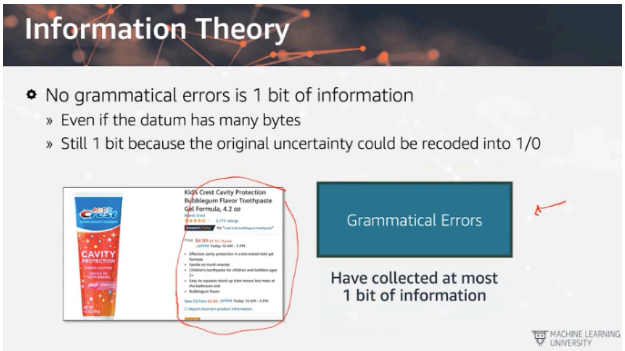


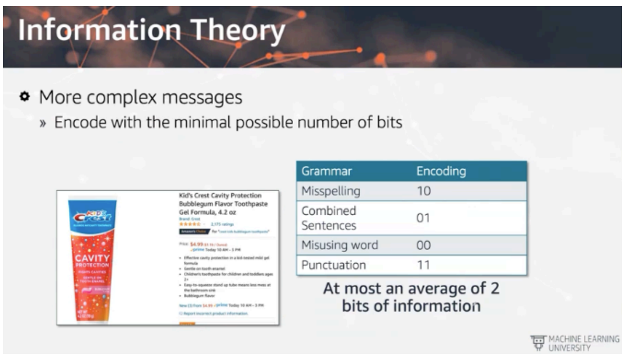
Vemos que la variable grammar, aporta mucha más información sobre si el objeto es fraudulento que solo tener una foto

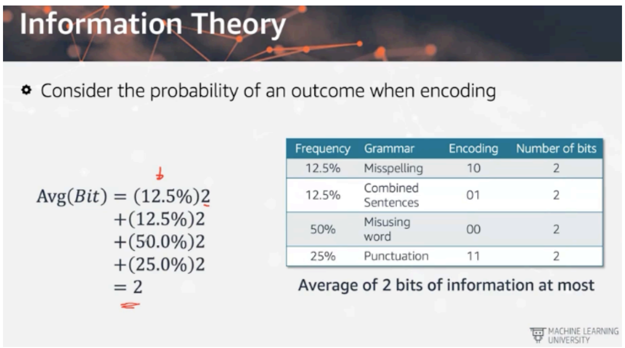




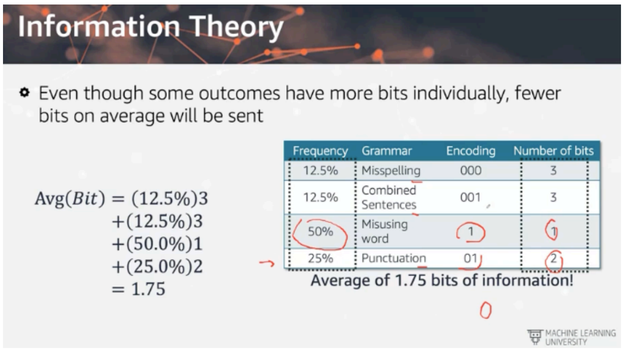






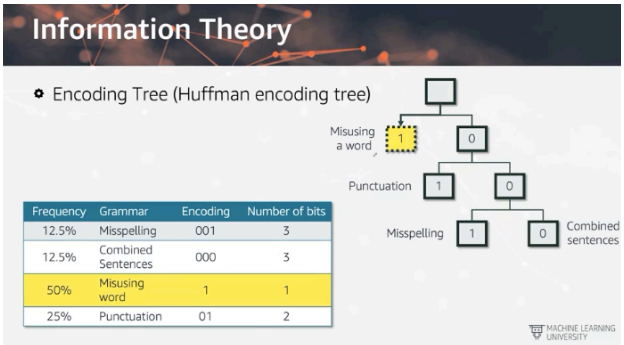


Lo que se intenta es reducir al máximo enviar toda la información lo más compacta posible (hoy en día el volumen ya no es un problema). Asignamos a los que tienen mayor peso a los que tienen más frecuencia



Frequency	Grammar	Encoding	Number of bits
12.5%	Misspelling	000	3
12.5%	Combined Sentences	001	3
50%	Misusing word	1	1
25%	Punctuation	01	2

Lo que se acaba formando es un encoding tree, se debe poner siempre los más probables los primeros para ir bajando si no se cumplen.



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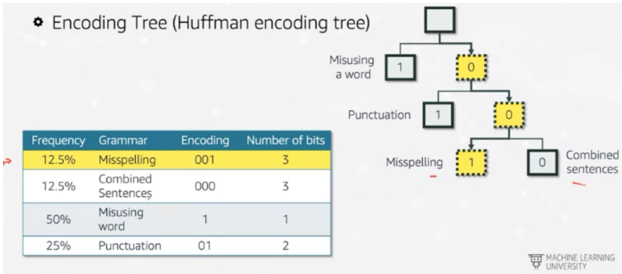
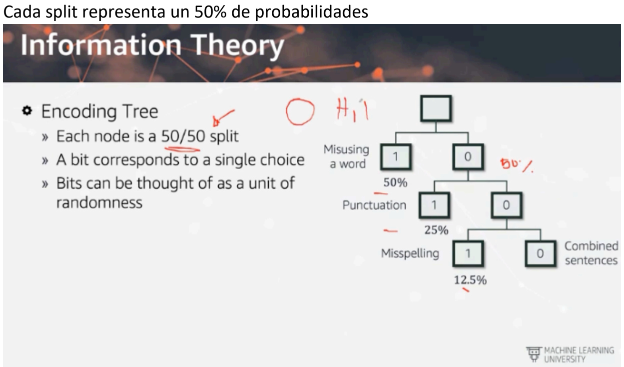
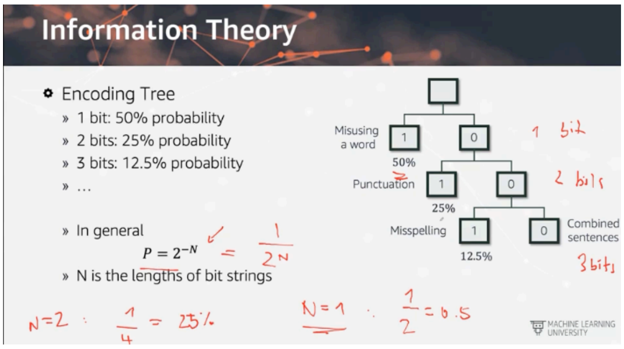


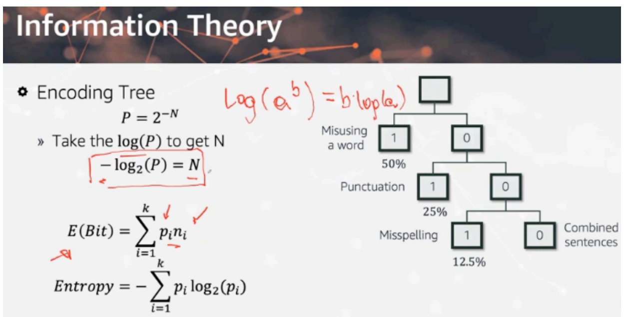
Frequency	Grammar	Encoding	Number of bits
12.5%	Misspelling	001	3
12.5%	Combined Sentences	000	3
50%	Misusing word	1	1
25%	Punctuation	01	2

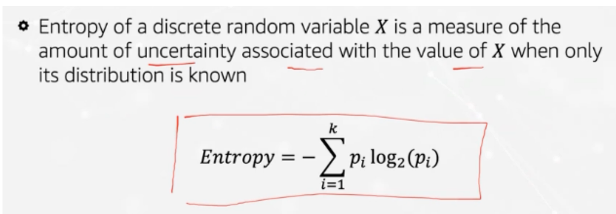
Cada split representa un 50% de probabilidades



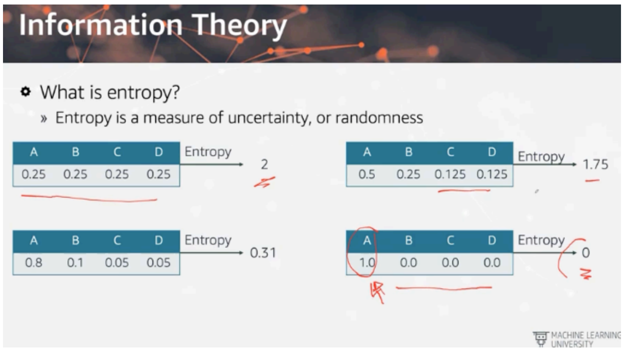


La entropia se utiliza para reducir la incertidumbre de los decision Trees.





SI toda la información se concentra en una sola posibilidad no hay ninguna entropia, Mientras que cuanto más repartido esté más incertidumbre tendremos.

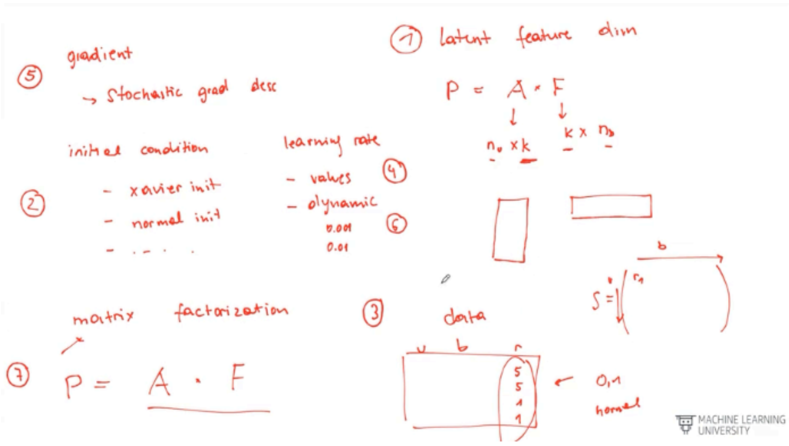


Relationship to machine learning

This is all well and good, but how does this relate to machine learning? Well the connections are many, but one of the clearest is in the evaluation of the performance of language models. What these try to do is to learn to generate text in some natural language, like English. The way it does this is that it looks at the text generated such far, and then tries to predict the next character (or potentially several characters or words). It then assigns a probability to each subsequent character. This can be fed into (a slightly modified) definition of entropy so you can understand how many bits of randomness remain after the model tries to make predictions. For many tasks this is the standard unit of measurement, with current top modeling methods getting down to about 0.98 bits per character (BPC).

It doesn't end there, entropy is also key to many techniques in ML such as random forests, and is a cornerstone of understanding how to measure the degree of error a model makes.

Métodos que se pueden aplicar para mejorar nuestro sistema de recomendaciones de lo más sencillo (1) a lo más complejo (7)



Dynamic - cambiar el Learning rate en cada pasada

Normalize the data Users en una dimensión y rating en la otra dimiensión, scale rating en 0,1 or normal.